

Experiment 4 Report: Final Difficulty Tournament

1. Objective

Experiment 4 evaluates the final Easy / Medium / Hard AI configurations in a larger tournament. Unlike Experiment 3, which screened candidate configurations, this experiment uses only the selected three difficulty levels.

The goal is to check whether the final difficulty ladder behaves as expected:

- Hard should be strongest overall.
- Medium should be clearly stronger than Easy.
- Easy should remain fast and weaker.
- Runtime should remain acceptable for the intended difficulty level.

The final cleaned result directory is:

```
experiments/experiment4_difficulty_tournament/result12
```

2. Final AI Configurations

Difficulty	Profile	Search	Evaluation	Depth	Candidate radius
Easy	A1	Alpha-Beta	Eval A	1	2
Medium	A3	Alpha-Beta + ordering	Eval A	3	2
Hard	B2	Alpha-Beta + ordering	Eval B	2	3

The final settings reflect the earlier experiments:

- Experiment 1 showed that Alpha-Beta pruning is useful.
- Experiment 2 compared evaluation functions and showed Eval B is a strong practical choice.
- Experiment 3 screened candidate configurations and identified A1, A3, and B2 as the final Easy / Medium / Hard set at that time.

3. Tournament Protocol

The tournament uses:

- 3 difficulty AIs;
- 3 pairings: Easy vs Medium, Easy vs Hard, Medium vs Hard;
- 200 games per pair;
- 600 games total;
- fixed opening positions;
- max moves: 50;
- draw if max moves is reached.

The tournament uses fixed openings rather than one empty-board start only. This reduces overfitting to a single deterministic game path and gives each pair a broader set of starting conditions.

4. Overall Results

Overall outcomes across 600 games:

Black wins: 320
White wins: 270
Draws: 10
Max-move draws: 10
Average game length: 17.57 moves

The low draw count indicates that most games ended before the 50-move cap. Black still has a small advantage, but white also wins many games, so the result is not dominated by first-player wins.

5. Ranking Table

AI	Profile	Evaluation	Depth	Games	Wins	Losses	Draws	Win rate	Score rate	Avg time / move ms	Avg nodes / move	Black win rate	White win rate
Hard	B2	Eval B	2	400	280	110	10	0.7	0.7125	3175.6271	431.85	0.75	0.65
Medium	A3	Eval A	3	400	250	140	10	0.625	0.6375	6228.0128	5356.2	0.65	0.6
Easy	A1	Eval A	1	400	60	340	0	0.15	0.15	17.6245	76.49	0.2	0.1

The ranking is ordered by score rate.

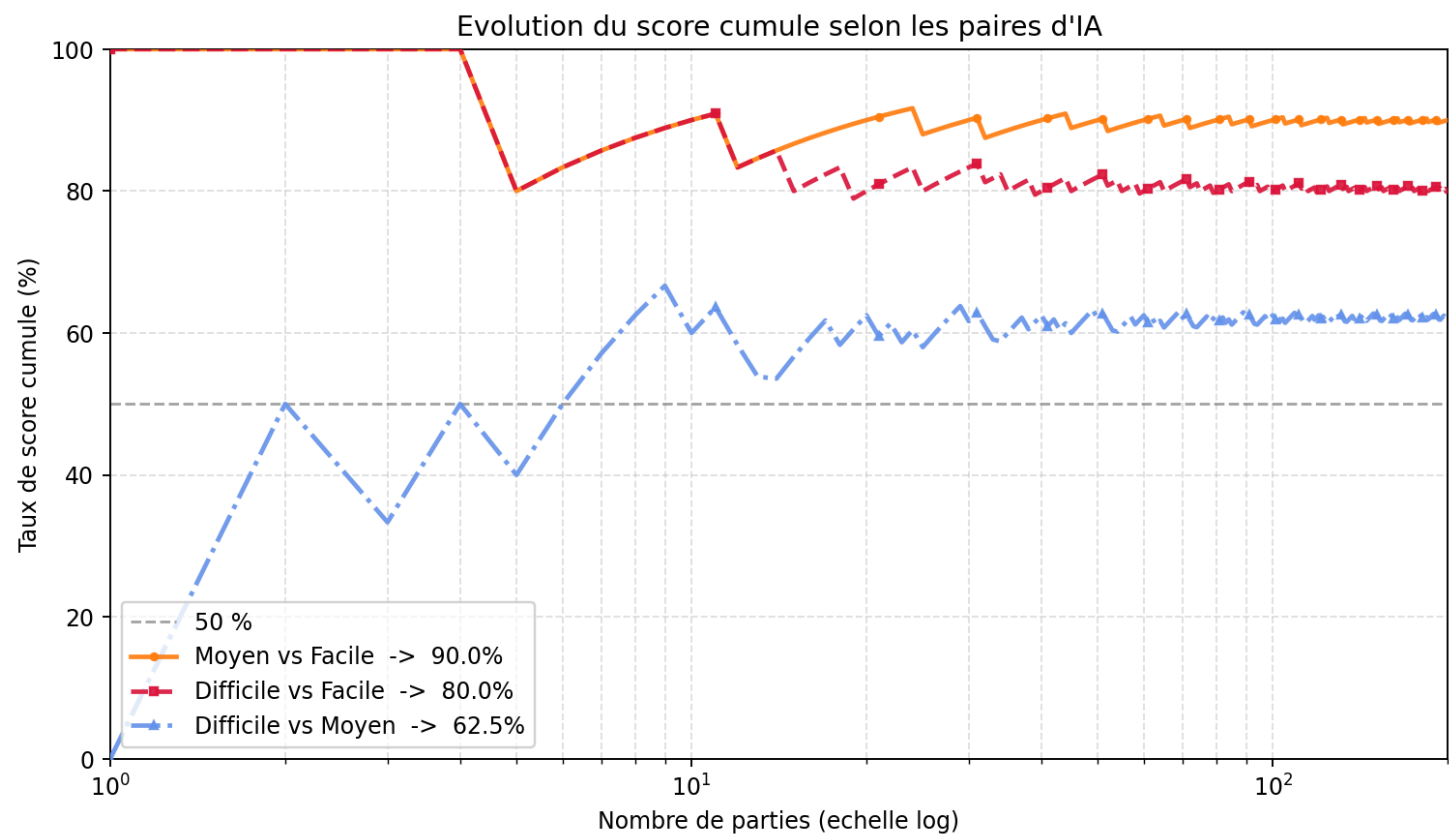
6. Pairwise Results

Pair	Games	AI 1 wins	AI 2 wins	Draws	AI 1 score rate	AI 2 score rate	Interpretation
Easy vs Hard	200	40	160	0	0.2	0.8	Hard clearly outperforms Easy.
Easy vs Medium	200	20	180	0	0.1	0.9	Medium clearly outperforms Easy.
Medium vs Hard	200	70	120	10	0.375	0.625	Hard is stronger than Medium, but the gap is smaller.

The pairwise table supports the intended difficulty ordering:

Hard > Medium > Easy

7. Cumulative Score Trend

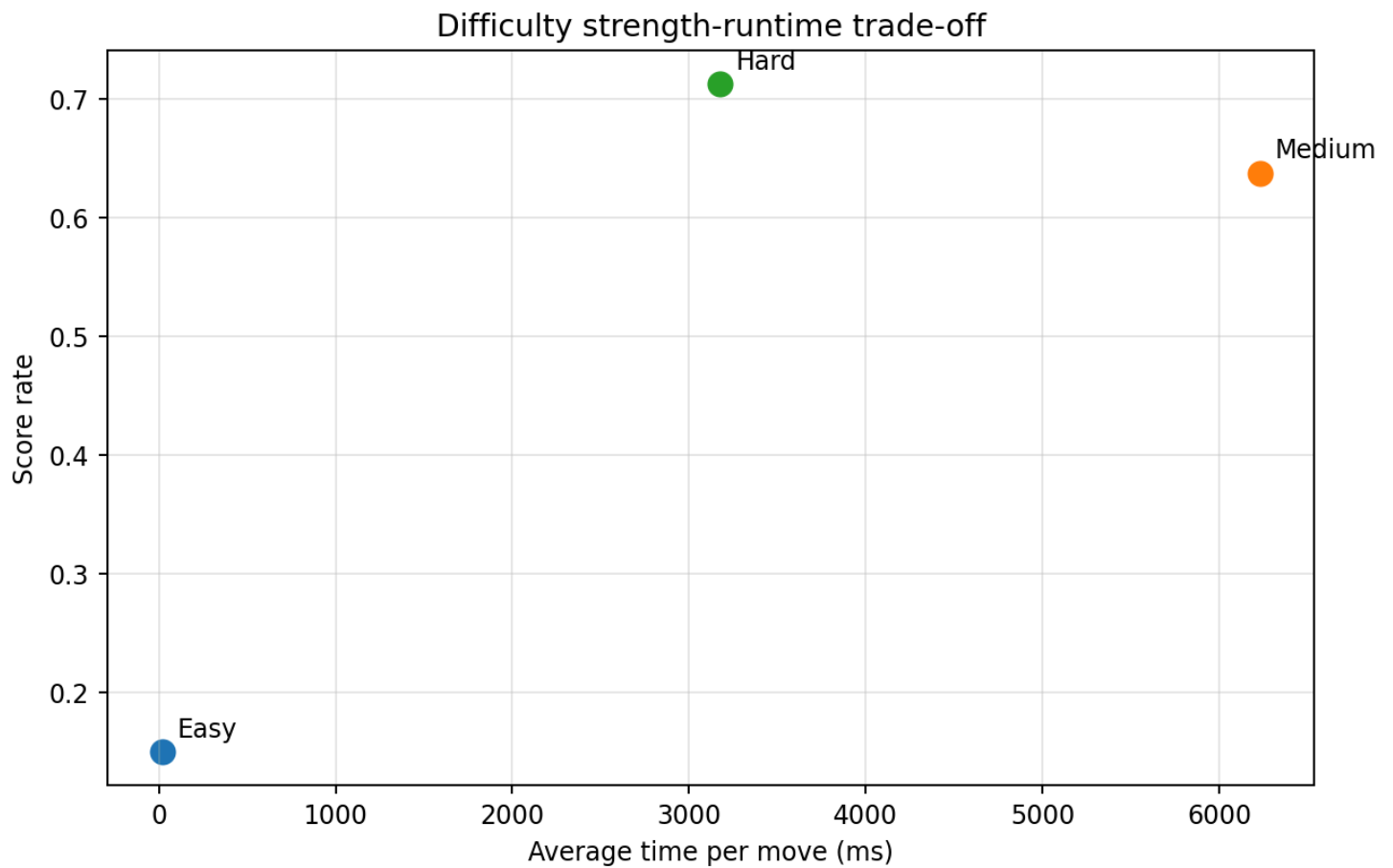


This plot tracks the cumulative score rate over games for each pairing. It is useful because it shows whether the final result stabilizes as more games are played.

The curves support the final ordering:

- Medium stays clearly ahead of Easy.
- Hard stays clearly ahead of Easy.
- Hard also leads Medium, though with a smaller margin.

8. Strength-Runtime Trade-Off



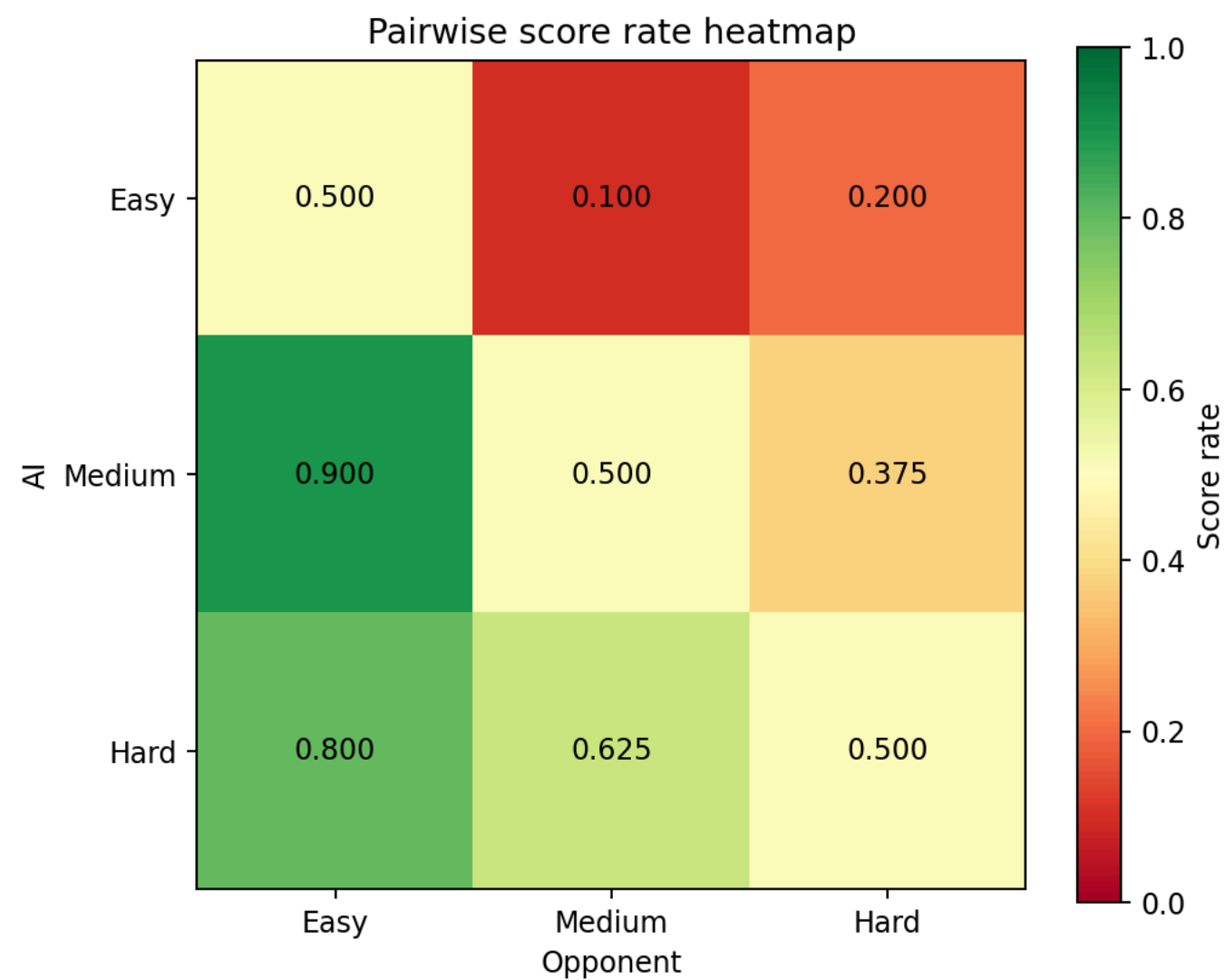
This plot compares score rate against average time per move.

Key observation:

- Easy is very fast and weak.
- Medium is stronger than Easy but is the slowest configuration.
- Hard is strongest and faster than Medium.

This is an important result: Hard is not simply the slowest AI. Hard uses Eval B at depth 2 with a wider candidate radius, while Medium uses Eval A at depth 3. The result suggests that better evaluation and candidate settings can outperform deeper search with lower runtime.

9. Pairwise Score Heatmap

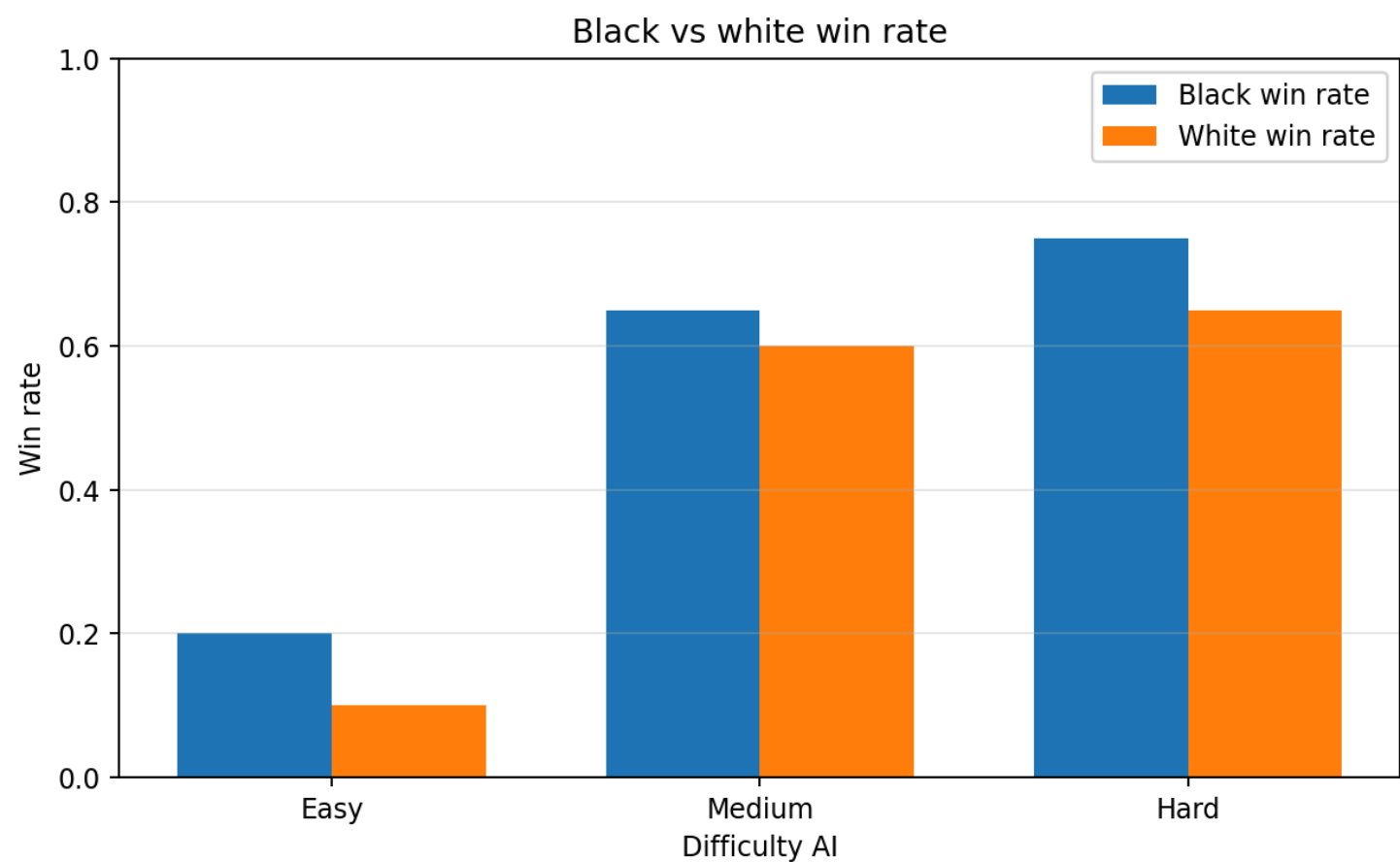


The heatmap gives a compact view of pairwise strength. Values above 0.5 mean the row AI scores better than the column AI.

The heatmap confirms:

- Easy scores poorly against both Medium and Hard.
- Medium strongly beats Easy.
- Hard beats both Easy and Medium.

10. First-Player Analysis



The color win-rate plot compares black and white performance for each AI.

Current results:

```
Easy black win rate:  0.20
Easy white win rate:  0.10
Medium black win rate: 0.65
Medium white win rate: 0.60
Hard black win rate:  0.75
Hard white win rate:  0.65
```

Black performs slightly better for all three difficulty levels, which is expected in Gomoku-like games. However, the gap is not large enough to overturn the main ordering.

11. Interpretation

The final difficulty ladder is valid according to the tournament results.

Easy is intentionally weak and fast. It loses heavily to both Medium and Hard, which makes it appropriate as a beginner-level opponent.

Medium is much stronger than Easy, but it is slower than Hard because it uses Eval A at depth 3. This makes Medium a useful middle level in playing strength, but not the most efficient configuration.

Hard has the highest score rate and beats Medium directly. It also runs faster than Medium on average, showing that the chosen Hard configuration has the best strength-runtime trade-off among the final three.

12. Conclusion

Experiment 4 supports the final difficulty setup:

```
Easy   = A1
Medium = A3
Hard   = B2
```

The tournament shows clear separation between Easy and the stronger levels, and Hard is the strongest final configuration. The Medium-Hard gap is smaller than the Easy-Medium gap, but Hard still wins the direct matchup and has the highest overall score rate.

13. How to Reproduce

Run the tournament:

```
python experiments/experiment4_difficulty_tournament/tournament.py
```

Regenerate plots:

```
python experiments/experiment4_difficulty_tournament/plot_difficulty_tournament.py
```

Final outputs are saved in:

```
experiments/experiment4_difficulty_tournament/result12
```